# Xiangjun (Ethan) Fu

xjf@seas.upenn.edu | ethanfxj@outlook.com | (858) 539-5848 | Philadelphia, PA

## EDUCATION

**University of Pennsylvania** — M.S.E. in Computer & Information Science **Aug 2025 – May 2027**
Cumulative GPA: 3.95 / 4.0

**University of California, San Diego** — B.S. Mathematics–Computer Science **Sep 2021 – Jun 2025**
Cumulative GPA: 3.9 / 4.0 | Provost Honors

**Relevant Coursework:** Statistical NLP, Deep Learning, Deep Generative Modeling, Computer Vision, Reinforcement Learning, Recommender System & Web Mining, Machine Learning, Discrete Mathematics & Graph Theory, Probability & Statistics.

## PUBLICATIONS

** equal contribution*

- Yupeng Hou*, Jiacheng Li*, **Xiangjun Fu***, Zhankui He, An Yan, Xiusi Chen, Julian McAuley. "Bridging Language and Items for Retrieval and Recommendation: Benchmarking LLMs as Semantic Encoders." **ACL 2026 Main Conference – Oral**.
- **Xiangjun Fu***, Guancheng Tu*, Suhao Yu, Yao Tang, Haoqiang Kang, Lianhui Qin, Yizhe Zhang, Jiatao Gu. “Latent Reasoning with Normalizing Flows.” *Under review at ICLR 2027*; **COLM 2026 on Efficient Reasoning**. arXiv:2606.06447.
- Cheng Yang, **Xiangjun Fu**, Jiatao Gu. “Decoupled Reading: Letting Demonstration-Conditioned Policies Follow Longer Compositions.” *Under review at CVPR 2027*; **CoRL 2026 on Compositional Generalization for Real-World Robot Learning**.
- Xinhua Wang, **Xiangjun Fu**, Caibo Feng, Chunxiao Liu. "Larger Hausdorff Dimension in Scanning Pattern Facilitates Mamba-Based Methods in Low-Light Image Enhancement." *Under Review*.

## RESEARCH EXPERIENCE

**GMLR Lab, University of Pennsylvania** — *Advised by Prof. Jiatao Gu* **Aug 2025 – Present**

- Co-led **Latent Reasoning with Normalizing Flows** (co-first author), embedding an autoregressive **normalizing flow** in the causal stream of **Qwen3-8B** for continuous latent chain-of-thought reasoning with exact, tractable likelihoods.
- Fine-tuned an **LLM-based VAE** to compress explicit reasoning trajectories into structured latents, then trained prompt-conditioned flows with a **unified flow + cross-entropy objective** on 732K code-reasoning samples (8×B200, FSDP).
- Leveraged **step-wise latent log-likelihoods** for density-based trajectory reranking and reinforcement learning over continuous reasoning paths; extended evaluation to math reasoning (AIME, MATH500, OlympiadBench).
- Co-developed **Decoupled Reading**, enabling demonstration-conditioned robot manipulation policies to follow longer skill compositions than those seen in training.

**McAuley Lab, UC San Diego** — *Advised by Prof. Julian McAuley & Dr. Yupeng Hou* **Dec 2024 – Dec 2025**

- Co-first author on a large-scale benchmark evaluating **LLMs as semantic encoders** across retrieval and recommendation tasks.
- Built a public Amazon dataset with **2M+ training and 250K+ testing instances** spanning collaborative filtering, sequential recommendation, and product search.
- Evaluated embeddings from **SimCSE, E5-Mistral-7B, Qwen-7B, Gemini, and OpenAI models**, analyzing their effectiveness across recommendation and retrieval settings.

**Shang Data Lab, UC San Diego** — *Advised by Prof. Jingbo Shang & PhD candidate Letian Peng* **Sep 2023 – Jun 2024**

- Developed a **weakly supervised text classification** framework using KL-divergence alignment between token-level representations and class-prompt distributions, improving **AG News accuracy from 78% to 93%**.
- Implemented and evaluated the framework in **PyTorch** across BERT, RoBERTa, and Gemma on AG News and Rotten Tomatoes.

## TEACHING & COMMUNITY

**Academic Achievement Hub, UC San Diego** — *Supplemental Instruction Leader* **Jul 2024 – Jun 2025**

- Led weekly Physics and Calculus workshops for 50+ students (20 hrs/week) with active-learning lesson plans; provided one-on-one mentorship in coordination with faculty.

## SKILLS

**Programming:** Python, C/C++, Java, SQL, Bash

**ML & NLP:** PyTorch, Hugging Face Transformers, LLM fine-tuning (LoRA, full fine-tuning), normalizing flows, flow matching/diffusion, VAEs, embedding models (BERT, RoBERTa, Mistral, Qwen), OpenAI / Gemini / Claude APIs

**Libraries & Tools:** NumPy, Pandas, Matplotlib, Jupyter, Git, Vim, gdb, SLURM / HPC clusters, conda

**Spoken Languages:** Fluent in English and Mandarin; conversational Cantonese and Spanish